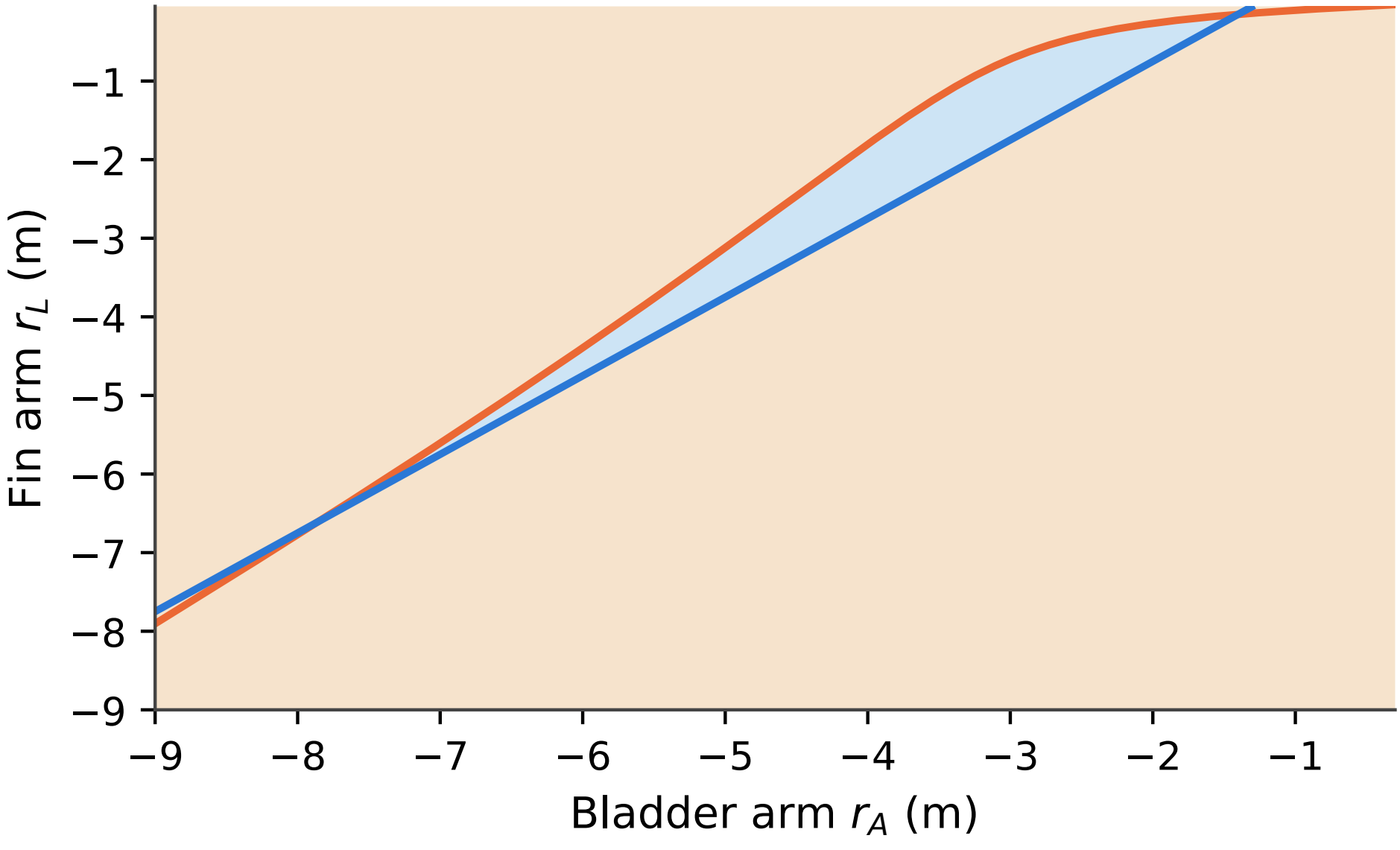

stable equilibrium

unstable equilibrium

$a_4 = 0$: constant term of the characteristic polynomial

$B = 0$: highest Hurwitz minor Δ_3